

PLACEHOLDER

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Abstract—Draft. UIE support object detection underwater, however, in some scenario, an enhancer that helps one image may not help another. We study selective enhancement: for each image, a lightweight ResNet18 selector chooses between the original image and the outputs of two pretrained enhancers, SCNet and Semi-UIR, before the image is passed to a YOLO11n detector.

Selector labels are produced with a three-fold out-of-fold protocol based on per-image average precision. On the RUOD benchmark, the selector reaches mAP, compared with for the original images and for the best fixed enhancer, and closes of the gap to an oracle selector.

Index Terms—underwater image enhancement, underwater object detection, selective enhancement, YOLO11, RUOD

I. INTRODUCTION

Underwater image enhancement is an important problem in computer vision. Li et al. proposed an underwater image enhancement benchmark dataset and discussed several challenges in this field [1].

REFERENCES

- [1] Chongyi Li, Chunle Guo, Wenqi Ren, et al. An underwater image enhancement benchmark dataset and beyond. *IEEE Transactions on Image Processing*, 29:4376–4389, 2020.